State Grid Customer Service Center
Tianjin 300300, China

6 November 2026

Editorial Office, *Computers* (MDPI)

Dear Editors,

We submit the manuscript **"HARK: Human-Accountable, Authority-Typed Review Knowledge for Evidence-Bearing Document Review"** for consideration as an *Article* in *Computers*, Section *AI-Driven Innovations*.

Problem. Organisations increasingly pre-screen documents with language models before an accountable expert signs off, yet the expert's review knowledge enters such systems as keyword rules, prompt checklists, or retrieval over past reviews. Each route loses a property that high-stakes document review needs: a finding must trace to a stated criterion and to a location in the document; the permission to resolve a criterion must match how deterministically that criterion can actually be checked; the knowledge must accumulate from use; and that accumulation must remain inspectable and revocable.

What the paper does. HARK is an architecture and an executed system that represent review criteria as rules carrying an *operational authority type*. Deterministic verifiers own `hard` checks and may produce absence records that a model is never allowed to assert; `hybrid` rules let verifiers locate candidate spans while the model judges only inside them; `soft` items are referral-only and never counted. A positional evidence gate checks submitted quotations against verifier-provided spans, a deterministic policy derives the suggested verdict from the confirmed set, and decision records persist the expert's dispositions. The manuscript reports the architecture, the implementation, the rule package and a retrospective evaluation, and it separates implemented behaviour from three target contracts that the evaluated snapshot does **not** enforce (authenticated human approval, a complete effective-authority ceiling, and content-addressed configuration recovery). Technical solution proposals are the empirical instantiation; assurance cases, compliance dossiers and architecture designs are potential applications, not domains validated here.

Evidence. On a fixed developmental regression corpus of 100 synthetic proposals with 253 planted defects, the combined configuration reaches micro-F1 0.798 (TP/FP/FN = 168/0/85) against 0.222 for a keyword baseline of the same policy (47/124/206), with 85% policy-verdict agreement (κ = 0.763). The result is deliberately bounded: 68 of the 85 misses are referred to pending review rather than silently passed, and the stored results do not show that a reviewer resolved them; 15 of 70 policy-non-Pass documents are under-called, including four of 23 expected Fail documents that receive a suggested Pass; gate acceptance is 42/53 and measures quotation length and overlap, not semantic support. Only 10 of the 28 shipped rules proved deterministically checkable and all four blocker rules are `hybrid` — a finding about the legacy checklist, reported as such. Operationally, the organisation's workflow handles more than 100 proposals per month and has accumulated thousands of records over several years. The instrumented evidence is five development-team runs on three internal proposals, reported as workflow evidence only, not as independent expert validation.

Fit. *Computers* publishes complete computational systems with reproducible evaluation, and this manuscript contributes exactly that: an executable review architecture, its data model and interfaces, a mechanism-level comparison, and an explicit account of what the evidence does and does not support. The Section *AI-Driven Innovations* matches the contribution — the representation and governance of review knowledge for AI-assisted expert work — and the Topic *AI Agents: Progress, Architecture, and Applications* (alternative: *AI-Driven Software Engineering and Next-Generation Web Services*).

Reproducibility and declarations. The extension package, the 28-rule package, the corpus generator, the 100-proposal corpus with planted-defect truth, the ablation runner and the published runs are released under MIT at `https://github.com/yyyyyyzt/HARK-Human-Gated-Authority-Typed-Review-Knowledge` (the repository path retains the project's original name) and archived at Zenodo (DOI 10.5281/zenodo.23955205); the manuscript states that the archived source (commit cbe1d33) precedes the cited snapshot by three commits that add DOI wiring, quotation checks and release notes, with the data byte-identical. The manuscript is original, is not under consideration elsewhere, and all authors have approved this submission. The authors declare no conflict of interest and no external funding; all authors are affiliated with the organisation whose review workflow supplies the internal pilot, as disclosed in the manuscript. The use of a generative AI assistant for language editing and manuscript preparation, without generating any experimental result, is disclosed in the manuscript.

Yours sincerely,

Zhitong Yang (corresponding author), on behalf of all authors State Grid Customer Service Center, Tianjin 300300, China yyyyyyzt@tju.edu.cn · ORCID 0009-0002-1803-1529